CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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I. INTRODUCTION

The spine typically has seven cervical, twelve thoracic, and five lumbar vertebrae. A vertebra is classified by where it falls in that sequence. Transitional variants sit at the two ends of the lumbar column, with the thoracolumbar transitional vertebra (TLTV) above and the lumbosacral transitional vertebra (LSTV) below. Both transitional variants disturb identification by changing what the border vertebra and its ribs *look like*. For example, a thoracolumbar border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly wedged body, a reduced disc beneath it and hypoplastic facet joints, or separated from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc[1]. In addition, they change how many vertebrae each region holds: eleven or thirteen thoracic and four or six lumbar.

Together, these transitional variants leave the vertebra at each border without a definitive name. The one formal nomenclature at the lumbosacral junction, Castellvi's, grades the morphology of the transverse process and says nothing about whether the segment is L5 or S1[1,2]; every imaging landmark proposed for numbering has proved unreliable in transitional cases[3–5]; and even a count from C2 yields a position, which a convention then converts to a name. Each of the readings above yields the same count (Fig. 2), and the morphology that would separate them is not always decisive to human readers. Whether it differs enough for a trained classifier to separate them is the question this release aims to answer. The gold standard for numeration in transitional anatomy is whole-spine imaging, counting caudally from C2[6], which determines how many vertebrae and how many rib pairs the column holds. No spine-limited acquisition can supply either count. LSTV is common, reported at 10–29% in the general population[7] and in 16.3% of a whole-spine CT series[8], and wrong-level surgery is its most serious consequence.

Existing public collections cannot address this, and not for lack of size (Table I). Prior spine datasets omit the pelvis and pelvic datasets do not number the vertebrae. TotalSegmentator[9] has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and VerSe[10], the one collection with an L6 class, stops at the sacrum and carries no ribs. As such, none of these collections can record a spine with six lumbar vertebrae together with both of its borders, and a segmenter trained on them must shift the whole column by a

level or absorb a vertebra into its neighbor when it encounters an enumeration anomaly. This problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS[11] raised that to 99%[12]. Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so what exists is a capability, not a dataset.

On the intraoperative side, LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT's vertebral labels onto it[13–15]. Those labels, however, are placed by hand and verified by the surgeon[13], so the method assumes a correctly labeled CT and does not supply one itself. As such, a dataset carrying the anomaly classes is vital for automating that step.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and classifies each structure an enumeration anomaly produces, including a sixth lumbar vertebra and lumbar ribs.

The project began with the question whether a vertebra can be named from the anatomy on the images a lumbar surgical case is actually planned on. By guideline those images are lumbar-only[16–19] and whole-spine coverage is reserved for trauma with an identified injury[20,21] and for deformity radiographs. An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it captures that span with the lowest ribs and the whole pelvis, while lacking C2. Every record comes from one prospective trial protocol, ACRIN 6664[22,23], which scanned 802 patients aged 50 and over, supine and prone, on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge, it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol. The comparators, by contrast, pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table I). With the protocol for this study fixed, scanner effects on a model become measurable rather than confounded. The downside is the restriction to an abdominal field of view (FOV), in which only the lowest thoracic levels are present (Sec. V).

CTPelvic1K[24] and CTSpine1K[25] each annotated the COLONOG collection under radiologist supervision for the same patients, one the pelvis and the other the vertebrae. The two had remained separate, although joining them needs no new imaging and no new radiologist. It was not a file merge, because each annotation had to be traced back to the series it was drawn on (Sec. II.A). Once completed, that merge yielded a combined spine-and-pelvis

frame for 802 patients, and the bones neither set had, ribs, femora and hardware, could be annotated against it rather than from scratch.

I.A. Vertebra labeling when the scan cannot settle the count

No scan in this dataset contains C2, a property of abdominal imaging rather than of the annotation, so the conventional count cannot be performed. A thirteenth thoracic vertebra and an L1 with a lumbar rib are different phenotypes, separated in cadaveric classifications of the thoracolumbar junction on quantitative shape rather than on position[26,27]: the first is a thirteenth rib-bearing vertebra above five lumbar vertebrae, 25 presacral in all; the second is a rudimentary rib on a lumbar-type L1 in a column with the standard 24. Counting distinguishes neither, because four lumbar vertebrae can mean a lumbar rib, a cranially shifted T13 or an L5 assimilated to the sacrum, and six can mean an aplastic twelfth rib or a sixth lumbar vertebra. A *stump rib* is a hypoplastic rib, a key indicator of a thoracolumbar transitional vertebra.

In this release, every vertebra carries the identifier its radiologist-sourced annotation assigned as the ground truth, corrected where the source was wrong. What is measured, however, is the bone and not its place in a sequence: body height, end-plate width, canal width and depth, pedicle width, wedge ratio, transverse-process span and its distance from the ala, and the length of the lowest rib as a fraction of the one above it. Each describes a vertebra in its own right, and none shifts with the name a reader gives the junction, so cases that disagree about the name remain comparable, and a criterion built on them is not a count under another name.

A vertebra can be named from its own morphology, and that a level has a characteristic shape is long established: pedicle width and height change systematically from the thoracic to the lumbar spine,[28] vertebral body, end-plate and canal dimensions differ by level,[29–31] and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.[1,32] If those differences hold at the boundary, the phenotype is decidable locally, and on this dataset it is. Separating T12 from L1 on shape alone gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. This was calculated using a logistic regression on five released per-level measures and six ratios between them. Each measure was divided by the patient's median across their levels, so

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it[34].

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K[25]	1,005	■	■							
CTPelvic1K[24]	1,184			■	■	■				
VerSe[10]	374	■	■							□
RibSeg v2[35]	660						■	■		
TotalSegmentator[9]	1,204	■		■	■	■	■		■	
CTSpinoPelvic1K	802	■	■	■		■	■	■	■	■

that size, which separates thoracic from lumbar trivially and says nothing at the junction, is removed. Folds are grouped by case and the curve is taken on pooled out-of-fold scores (`test_morphometric_separability.py` in the released code). Schinz et al. reach the same conclusion from the other side: on 1,242 whole-thoracolumbar CTs a shape-based labeling of the junction matched nerve morphology in every case, against 92.6–97.2% for counting- and rib-based rules.[33]

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

CTSpine1K's 1,005 volumes and CTPelvic1K's 1,184 come from four and seven collections of CT obtained for different indications. Only the Cancer Imaging Archive (TCIA) CT colonography collection (COLONOG)[22,23] is common to both, and it is the largest in each. Every COLONOG patient was scanned prone and supine at a minimum, with 12 exceptions among the 825 patients the sources drew on. CTSpine1K states that one position was chosen at random by a released script; that script, since deleted but preserved in the repository's history, writes every series holding twelve or more files to the same filename, so the volume retained is whichever series the directory walk reached last. No code was released with CTPelvic1K. Consistent with two independent choices, the sources annotated the same series for 342 of the 693 patients both annotated (49.4%), and the released records mix positions and reconstructions accordingly: 422 supine, 377 prone, three decubitus; 562 at 1.0 mm, 232 at 1.25 mm and eight at other slice thicknesses.

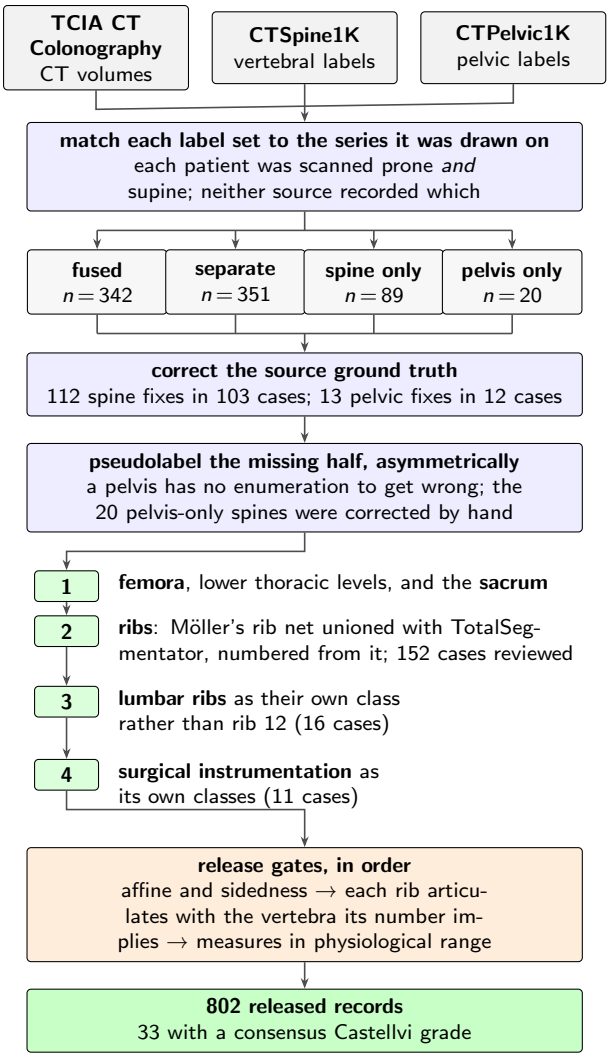


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

Recovering the archived series each annotation was drawn on is nontrivial because the DICOM attributes that identify an acquisition, the patient, study and series instance identifiers and the patient position,[36] have no counterpart in the NIfTI-1 header[37] and are discarded on conversion.[38] CTSpine1K's `Path.csv` names the TCIA directory each label was converted from, but it carries no series identifier, names no position for 217 of the 782 annotated records, and for seven indicates an acquisition other than the one whose bone the label overlaps best. CTPelvic1K released no imaging and encoded the mapping in its mask filenames, `dataset2_PatientID_SeriesNumber_undocumented_[tag_]mask_4label.nii.gz`. One of the 714 masks names a patient with no archived series; of the other 713, the first two fields resolve 702 to the series selected here by bone overlap, five to a `SeriesNumber` two series share, and six to one no series carries. The third field matches nothing, and the fourth, on 32 masks, mixes phenotype (`sacralization` 7, `semisacralization` 2, `hard_sacralization`

1) with difficulty, image-condition and unexplained tags in one string.

The crosswalk (Fig. 1) therefore resolved every annotation, not only the 27.7% with no stated position, by thresholding each of the patient's volumes at bone attenuation and keeping the series the mask overlaps most; where a source did name a series, 558 of the 565 CTSpine1K records naming a position and 702 of the 713 CTPelvic1K masks naming an archived patient agree with it. The result is 342 records whose spine and pelvic labels land on the same series (*fused*), 351 whose labels land on different series of one patient (*separate*), 89 with a spine label only and 20 with a pelvic label only. Turning a patient between prone and supine alters lumbar alignment and the seating of the pelvis against it[39], so a mask drawn on one series does not transfer rigidly to the other, and while a patient's two annotations lie on different scans no spinopelvic parameter can be calculated across them and the lumbosacral interface is not one geometry to learn from.

II.B. Densification, and why it is asymmetric

A rigid cross-registration of each pelvic mask onto the patient's other acquisition was attempted, and is released in the code base, but was found wanting. The current nnU-Net[40] framework has an `ignore` label that permitted training on pelvis-only, spine-only and fused masks together without degrading segmentation at the lumbosacral junction. A preliminary network was trained in five folds to pseudolabel the missing pelvises of every volume in the cohort, excluding those with fused masks, using out-of-fold inference so that no pelvis was pseudolabelled by a model that had seen it during training. That also permits scoring the model against the radiologist-annotated pelvises: on the manual labels of each held-out fold it reaches a sacrum Dice of 0.980 ± 0.003 with no fold below 0.977, and hip Dice of 0.970 ± 0.016 and 0.971 ± 0.015 with no fold below 0.953 (Table S1; per-fold values in `results/pseudolabel_dice/`). The pelvis is thus radiologist-derived on the 362 fused and pelvis-only records and pseudolabeled, close to radiologist quality, on the other 440 (Fig. 1). A pelvis carries no enumeration to get wrong, whereas a pseudolabelled spine must commit to a count, so the 20 pelvis-only spines were corrected by hand.

II.C. Ribs

No public rib dataset annotates the spine with a labeled lumbosacral or thoracolumbar transitional vertebra, and the two segmenters available for pseudolabeling both fail[9,41].

Möller's binary rib network[41] works very well when the whole rib is in the FOV, but fails on a rib even partly outside it, and since ribs grow more caudally angled toward the thoracolumbar junction, that failure puts a non-negligible amount of bone into the background. TotalSegmentator[9] segments ribs both partly and wholly inside the FOV and assigns each a numeric class, which a binary network cannot, but its ribs stop short of the costovertebral interface. That blocks the quality-control test confirming that a rib's class label matches the class of the vertebra it is incident on.

The gaps being complementary, the two were combined (Fig. 1): TotalSegmentator supplied the numbering and the partial-FOV ribs, Möller the ribs wholly inside it and reaching the vertebra. Their union is a pseudolabel of every rib in the dataset. Rather than review all of them, a quality-control pipeline triaged the likely errors (Table II) on two label-based checks: a rib bone carrying two or more labels, and a rib that did not reach the spine.

The triage identified 152 records. Every one was corrected by a reviewer and 149 finalised by a second read; the three not finalised are named in the release notes. Neither gate fires on any release record. What is left is seven ribs split inside the reconstructed field (five carrying a piece of a neighbour, two with the head apart), two offset ribs and eleven whose medial end stops short of a vertebra. Reviewers were medical students working through a student research consortium[42], trained against a written labeling protocol, with every correction attributed to its annotator and a save refused until the review checks passed. Since a rib is named for the vertebra it articulates with, the whole-side shifts left after review were renumbered against the vertebrae by that rule, so the release column is a consistency check on the rule and the pseudolabel column the independent measure of the pseudolabel and the reviewers.

II.D. Label scheme, and the phenotypes it expresses

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure

TABLE II Rib checks on the raw pseudolabel and on the release, the same code on both. The first two gated the review; the rest were measured on every record. Pairs are records / ribs; a piece is a second component of at least 50 voxels and 15% of the largest, and a rib that leaves the reconstructed field and re-enters it is not counted as split.

Check	Pseudolabel	Release
Two numbers on one bone	151 / 584	0 / 0
Rib detached from the spine	37 / 40	0 / 0
Rib in two pieces inside the field	119 / 280	6 / 7
Rib offset from its vertebra	24 / 170	0 / 2
No vertebra within reach	— / 38	— / 11
Rib on a lumbar body	14 / 21	0 / 0

takes a fixed identifier above that range — hips 30–31, femora 32–33, thirteen ribs per side 34–46 left and 47–59 right, lumbar ribs 60–61 and surgical hardware 62–68. Identifier 29 is retired and unused, an earlier scheme having given it to a separate S1 class, and is not reused because renumbering 30–68 to close the gap would rename the hips, femora and every rib under identifiers consumers are already keyed to. Rib 13 is the true rib of a T13 and is empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are labeled as the distinct phenotypes they are.

The sacrum is not sub-divided: it is the source annotation's own outer boundary, no S1 class is asserted, and the superior end-plate that sacral slope and pelvic incidence are measured from is fitted to the whole sacrum, isolated by the lowest lumbar body's footprint.

VerSe[10] carries L6, whose identifier this release keeps, so its L6 *is* VerSe's. A rib on a lumbar body, however, takes its own class, the one class here with no published counterpart. A scheme that numbers every rib 1–12 leaves the annotator two bad options: call it rib 12, which asserts that the vertebra beneath it is thoracic, or discard it. VerSe's T13 is a vertebra rather than a rib. This release records both: VerSe's T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib. No record in this dataset carries a T13. In each of the sixteen lumbar-rib cases the vertebra under the rib was named a lumbar body in the source annotation, and the rib class follows by rule, not judgement: the rib whose head sits on a lumbar body takes 60 or 61. Where a border vertebra could not be settled, the source reading stands rather than being

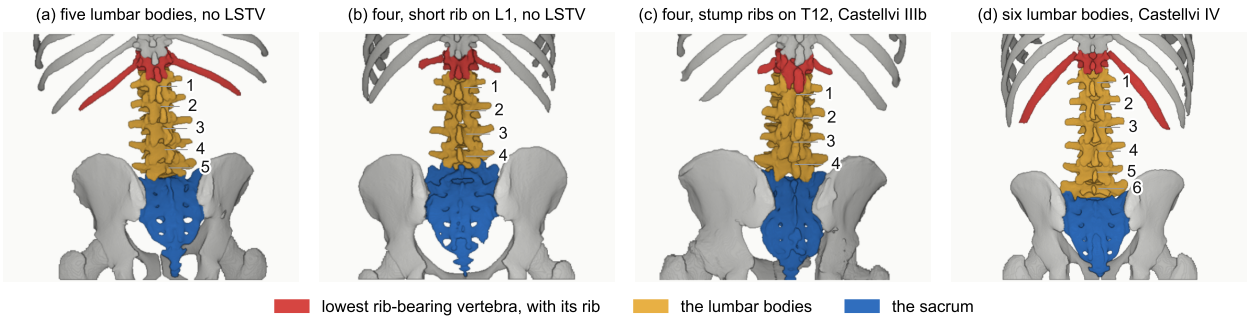


FIG. 2 Four phenotypes at the lumbar borders, aligned on the sacrum. Red, lowest rib-bearing vertebra and its rib; blue, sacrum; yellow, the lumbar bodies. Cases 0704, 0428, 0094, 0005, each the most ordinary member of its phenotype by rule. (b) and (c) present the same upper border, a short rib pair under a full pair, and were named differently.

overridden here. Of the sixteen, thirteen are bilateral and three unilateral, all on the right, too few to speak to side preference.

Figure 2 shows the four phenotypes the scheme has to express, and why a count does not settle any of them. (a) is the ordinary column, five lumbar bodies and no transitional vertebra. (b) and (c) present the same upper border — a rib pair of about 40 mm beneath a full pair, above four lumbar bodies — and were named differently: a lumbar rib on L1 in (b), stump twelfth ribs on T12 in (c). Nothing in the rib decided that, and the two populations barely separate on rib length alone, the sixteen ribs labeled lumbar running a median 45 mm and 0.32 of the rib above them against 38 mm and 0.28 for the 98 labeled stump twelfth ribs. What decided it was the convention of five lumbar vertebrae, which leaves exactly one place for the anomaly once the lower border is read, transitional in (c) and not in (b). (d) is the sixth lumbar body. A scheme carrying only ribs 1–12, five lumbar identifiers and no sacrum cannot write down (b), (c) and (d) as the distinct things they are; this one can, which is what makes them separable data rather than three readings of one count.

So the name a border vertebra carries in this release is a reader's, not a measurement's: no morphometric criterion was applied to overrule a reading, and a T12 and an L1 at this border are separated by the eye that read them. Every public collection follows that convention; it is stated here because the measurements released beside these labels are what a morphometric criterion would be built from, and some labels may not survive one. A user testing such a criterion should treat the border labels as expert reads, not as ground truth independent of the reader.

Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded, being the one instrumented record, pulled out of the normal review flow while its merged lumbar labels were rebuilt by hand and never returned to the readers. The source transitional label (pelvic where CTPelvic1K flagged one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and semi-sacralization in one, and is unremarkable in the ungraded record.

Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`, `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by counting identifiers 25 and 60–61.

II.E. Computational tools

Every step in this pipeline is performed by open-source code archived with the dataset. Segmentation of the femora and the per-level rib numbering used TotalSegmentator[9] (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller's[41] released weights, applied through the nnU-Net v2 framework[40]. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble applied out-of-fold, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (the repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image's grid and affine exactly.

Dimensions and spinopelvic parameters come from code released with the dataset. A sacral plate whose fitted normal lies more than 60° off the cranial axis is rejected and the case reported missing, since such a fit has found the near-vertical anterior face of the promontory, not the endplate. Gating on the geometry rather than on the value leaves unusual patients in. An open-source measurement toolkit is released separately (the URL identifies the authors and is on the title page) as a reusable implementation measuring in each vertebra's own frame. No open-source implementation computing these parameters from CT segmentations was found elsewhere.

A large language model (Claude, Anthropic) assisted with writing the analysis and figure code and with manuscript outlining and refinement. It was not used to generate, annotate or

interpret imaging data; every number reported here is computed by the released code from the released volumes. All content was independently reviewed, verified, and interpreted by the authors.

II.F. Validation

Validation has three levels. Geometry is checked first, because a measurement taken on a structure with a geometric fault still returns a plausible-looking number.

Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed.

The check compares *each sided pair separately*. Testing the ribs alone passed all 802 records, four of which carried a `left_hip` label on the patient's right side, and pooling every sided structure into one test still missed one of those four, because a large correctly-sided structure masks a smaller transposed one.

Each rib is matched to its nearest vertebra and compared against its corresponding vertebra. Of 11,560 ribs, 5,796 are not evaluable because the expected vertebra lies outside the FOV and 11 have no vertebra within the anchor distance. Of the 5,753 evaluable, 5,751 match and two (0.035%) are offset by one level. No case is misnumbered, which is a different call: a rib is offset when the vertebra it reaches is not the one its number implies, whereas a case is misnumbered only when three or more of its ribs are offset by the same amount, the signature of a side numbered from the wrong end. One or two isolated offsets are likelier a segmentation artefact, and that is what these are, single ribs in two records, both truncated by the edge of the field.

Of the 5,753 evaluable ribs, *zero* are assigned to a lumbar vertebra, despite 16 records carrying one, because a lumbar rib takes class 60 or 61 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be the labeling error this dataset exists to expose.

Pelvic incidence is the strongest of these checks because it is a morphological property rather than a posture (Table III, Fig. S2). This cohort's mean agrees to within a tenth of a degree with the automated supine CT series of Veilleux *et al.*[43] and sits below the standing

TABLE III Spinopelvic measures, mean ± SD over the 756 of 802 records passing the geometric gates, against two supine CT series that report means.[39,43] Standing values are 55.0, 41.0, 13.0 and 60.0°.[45] The end-plate is fitted to the whole sacrum, and the plate gate rejects 44 records. Pelvic incidence and tilt are right-skewed (0.25 and 0.33), so their medians—51.5 and 15.0°—sit below these means and the modal peak sits below both.

Measure	This dataset (supine, $n = 756$)	Veilleux ($n = 200$)	Hasegawa ($n = 24$)
Pelvic incidence	52.1 ± 11.1°	52.1	53.4
Sacral slope	36.6 ± 10.6°	36.5	34.1
Pelvic tilt	15.5 ± 7.9°	15.6	19.2
Lumbar lordosis	51.7 ± 13.3°	—	—
PI − LL mismatch	0.3 ± 10.3°	—	—

figure, the direction reported for subjects imaged both ways.[44] Sacral slope and pelvic tilt are postural, and read within a tenth of a degree of Veilleux's asymptomatic subjects. A second supine CT series sits within four degrees of this cohort on all three.[39] That residual is measured against 24 subjects, and both comparator series are supine, so posture cannot account for it; it tracks cohort and not construction.

II.G. Castellvi typing

Every record whose lumbar vertebra count differs from five carries a Castellvi grade[2] in the release, 33 cases typed I–IV with the a/b unilateral–bilateral qualifier. The grade and the count are *different axes*, the count being how many lumbar bodies the column holds. Grade IIIb occurs here at lumbar counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction they disagreed on, type II (articulation) against type III (bony fusion)[1], is the one that matters clinically, and four of the six disagreements were III against II.

II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect in CT and hard to interpret, and the difference matters here because **to a distance measurement an iatrogenic fusion is indistinguishable from a congenital one**: a cage-bridged interspace reads as "no gap" exactly as a fused transitional vertebra does.

A 2500 HU threshold inside a shell around the labeled skeleton proposes candidates, and every candidate is confirmed or rejected by manual review, because attenuation alone cannot separate an implant from contrast, calcification or reconstruction artifact — all of them saturate. Site and volume can: every confirmed implant measures 2,586 mm^3 or more and every rejection 1,768 mm^3 or less, and the confirmed object is typed from its site and shape. Identifiers 62–65 name spinal instrumentation (generic, cage, screw-and-rod, plate), to which this release adds **66 arthroplasty**, **67 sacroiliac screw** and **68 osteosynthesis**. Eleven records carry hardware: eight arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages, each shown in Fig. S1.

An implant lies inside the bone label that surrounds it, so the hardware class is taken *out* of that label rather than added beside it, reclaiming 1,538,852 voxels across the eleven records: a user measuring bone gets bone, and one measuring the implant gets the implant. In eight of the eleven, the femoral head that pelvic incidence and tilt are measured from is an implant.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM, which is what volumetric segmentation tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. Every record is a validation record in exactly one fold, so the release supports cross-validation but holds no records back from all five. A reader who wants one accuracy figure on data a model has never seen must set records aside before training, keep them out of every fold, and say which they were.

Stratification is on the transitional subtype rather than a binary LSTV flag, read from

the source transitional labels and not from the lumbar count. Across all 802 records the source label reads lumbarization in 14, sacralization in 17 and semi-sacralization in two; 18 records carry a sixth lumbar-type vertebra, nine only four lumbar identifiers, and 33 a consensus Castellvi grade.

The released file holds 768 normal, 15 sacralization, 15 lumbarization, two semi-sacralization and two ambiguous records. The ambiguous two are source-labeled sacralizations whose vertebral source reads lumbarization. The fifteenth lumbarization is record 0500, which the file lists as carrying an L6 although its released label has five lumbar vertebrae and no L6; why the splitter saw one is not known. It read an L6 on no other record, so one L6 record sits in the normal stratum. The two strata of 15 put three records in every validation fold; the two strata of two cannot appear in every fold at all, so a fold-level metric on the rare classes carries an error bar far wider than its own decimal places. A reader who wants folds balanced on the label volumes must regenerate them.

Per-field record counts across the manifest are given in Table S2.

The archive of record is Zenodo (https://doi.org/10.5281/zenodo.22139642); any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records: 393 female, 345 male, 11 carrying DICOM's *other* value and 53 with no sex field, with age present for 709 (median 59, range 50–89). All 802 are counted in totals; sex-stratified measures rest on the 738 recorded female or male.

The lowest-rib length ratio is bimodal, its lower mode lying below 0.33 of the rib above. Its lower mode marks a stump twelfth rib in 98 of the 789 records with a measurable pair (12.4%), and a lumbar rib is present in 16 (2.0%), against 12.6% and 1.5% in a whole-spine CT series that reports both[8]. That series also reports that the two rib anomalies travel with the lumbosacral border, stump twelfth ribs with sacralization and lumbar ribs with lumbarization. The 789 hold 14 of the 17 source-labeled sacralizations; stump ribs accompany four of those 14 and 94 of the other 775, and no lumbarization case carries a lumbar rib.

Vertebral bone density can be read off a study acquired for another indication without

further radiation.[46] L1 trabecular attenuation is reported here at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, so sacral morphology is measurable against the lumbar column rather than in isolation: the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves. No trajectory can be planned across a junction whose segments cannot be named, so the transitional layer serves that use rather than competing with it.

A proxy for the preoperative view. We are not aware of a public preoperative lumbar CT cohort, which makes this release the closest available stand-in for the view lumbar cases are planned on: 377 of its records were acquired prone, the position of posterior lumbar surgery, and it carries classes for the anomalies that make level identification fail. As a screening population at colonography dose, not a surgical series, it supports method development for level identification and instrumentation geometry rather than validation of a preoperative planning decision.

V.A. Three uses the released measurements support

Wrong-level spine surgery runs at roughly one in 3,110 procedures, and among the predominant contributing factors is the transitional anatomy this cohort was assembled around.[47,48] Per-level body height, canal width, end-plate width, transverse-process span and wedge ratio are released across all 802 records, so a model can be trained on the shape of a vertebra rather than on its position in a count.

The values a surgeon uses for endplate and canal dimensions and pedicle width come from cadaveric series of ten to thirty specimens[29–31] redrawn in the textbooks as one curve per level[49], and a standard error describes the mean, not the spread of individuals. The same measures are given here from 802 records with the distribution attached (Fig. 3),

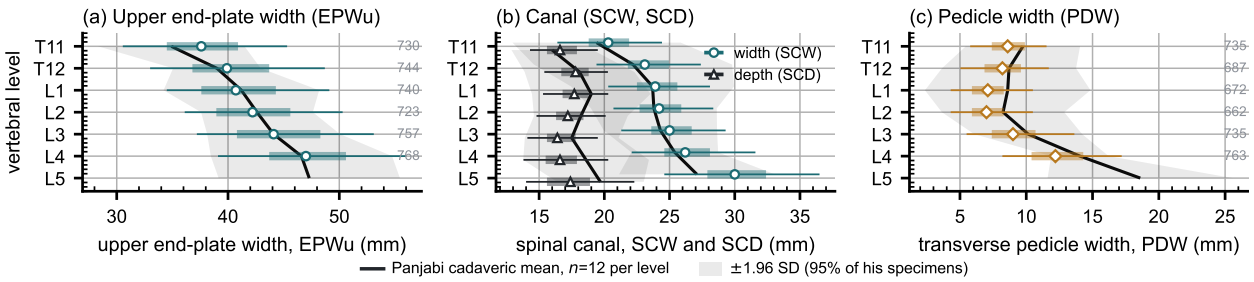


FIG. 3 Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile, n at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is Panjabi's cadaveric series[29,30], keyed in (a), its SD recovered from the reported SEM; T11–T12 EPWu is digitised from his Figure 4A.

reproducing the classical pedicle widening into the lower lumbar spine. Both layers of that figure are population intervals, not confidence intervals, and that is what makes their widths comparable: a confidence interval bounds a parameter, a population interval bounds individuals,[50] and at $n = 12$ the two differ by $\sqrt{12}$.

Patient-specific finite-element modeling for surgical alignment planning and implant design[51] needs spine, pelvis and femoral heads in one frame, which is provided by this release. 351 patients in this dataset give a within-patient postural change to check a prediction against.

Thoracic ground truth for this release is FOV limited and does not reach T1 (Fig. S3). In addition, the postural angles measured are supine. While pelvic incidence is not postural, it is modality-sensitive, and the measurements read below radiography for the same subjects.[44] End-plate and pedicle width are withheld at L5 because the transverse processes arise in front of the canal wall the body is cut at. Those measurements along with the spinopelvic angles are the provisional part of this release. The planned revision takes the vertebral body and the sacral plate from a substructure label,[11] which measures the body without the canal cut, and the version DOI carries it. The cohort is also one protocol of a colorectal screening population aged 50 and over, so its distributions should not be read as those of a surgical series.

Label strength varies by structure, and the release does not average over it. Vertebral labels derive from radiologist-supervised source annotations, corrected where wrong by trained medical-student reviewers working to a written protocol, who escalated any junction they

could not settle to a radiologist rather than overwrite it; pelvic labels on records that lacked one are pseudolabeled.

The ribs are pseudolabels, and their review was triaged rather than exhaustive: a rule selected the records most likely to be wrong and those were corrected, so what the release can promise is that the specific failures that rule detects are absent, not that every rib in every record has been inspected. The transitional labels come from two independent sources and are not uniformly adjudicated, which is why the measures reported here do not depend on the count; the Castellvi grades are a consensus of two radiology resident physicians. The sacrum is not sub-divided, so a user who needs a first sacral segment must delineate one.

Sacrum, hips and femora are present in all 802 records; one lacks T12, outside the FOV. Nine carry no L5 identifier because their source annotation counted four lumbar vertebrae; all nine are Castellvi IIIb, the fused segment is delineated with the sacrum, and the manifest names them. No shape-based classifier is attempted at the lumbosacral junction, where 15 records in each rare stratum leave three per fold; the 0.990 area under the curve above is the thoracolumbar boundary, not this one.

VI. DISCUSSION

VerSe[10,34] comes closest in intent to this dataset, but carries no sacral mask for the structure a Castellvi grade describes, which this release segments as L6 or as sacrum, and has neither the pelvis and femora a spinopelvic measurement needs nor prone acquisitions. What this release adds over TotalSegmentator[9] is the classes for anatomical anomalies, not necessarily more anatomy per scan.

Applying the released weights to VerSe would add the sacrum and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole cohort would test the co-occurrence of rib and lumbosacral anomalies at full power and supply the labels a lumbosacral classifier needs.

VII. CONCLUSION

CTSpinoPelvic1K places the radiologist-derived spine and pelvis and the pseudolabeled per-level ribs and femora on one validated coordinate frame in 802 records, giving the

anatomical anomalies that make lumbar numbering ambiguous classes of their own. Because every vertebra is measured in its own right rather than by its place in a sequence, a level can be named from the morphology in a field of view that does not allow the conventional count downward from C2. This release demonstrates that at the thoracolumbar border and leaves it open at the lumbosacral one, where it supplies the classes and the measurements to settle the question but not yet the number of transitional records a classifier would need.

SUPPORTING INFORMATION

Supporting information is available online and is not part of the article PDF. Fig. S1, the instrumentation gallery, one exemplar of each implant class drawn through bone. Fig. S2, the spinopelvic measures against standing reference bands. Fig. S3, the records carrying each vertebral level. Table S1, the pelvic pseudolabeller's held-out fold Dice. Table S2, data types and metadata and the records populating each. Table S3, the per-identifier label census.

DATA AVAILABILITY

The release described here is v11, permanently archived at https://doi.org/10.5281/zenodo.22797014; the concept DOI https://doi.org/10.5281/zenodo.22139642 resolves to the current version. The archive carries the loader, the label scheme and the quality-control tables under an open-source licence. The repository additionally carries the extraction and figure code, the reference table with a citation on every row, the values plotted in Fig. 3, and a recipe for running them in order. The repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version.

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CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors' institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

REFERENCES

[1]G. Konin and D. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, AJNR Am. J. Neuroradiol. **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.

[2]A. E. Castellvi, L. A. Goldstein, and D. P. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, Spine (Phila. Pa. 1976) **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.

[3]J. A. Carrino, P. D. Campbell, D. C. Lin, W. B. Morrison, M. E. Schweitzer, A. E. Flanders, J. Eng, and A. R. Vaccaro, Effect of spinal segment variants on numbering vertebral levels at lumbar MR imaging, Radiology **259**, 196–202 (2011), doi:10.1148/radiol.11081511.

[4]N. A. Farshad-Amacker, A. Aichmair, R. J. Herzog, and M. Farshad, Merits of different anatomical landmarks for correct numbering of the lumbar vertebrae in lumbosacral transitional anomalies, Eur. Spine J. **24**, 600–608 (2015), doi:10.1007/s00586-014-3573-7.

[5]D. Tureli, G. Ekinci, and F. Baltacioglu, Is any landmark reliable in vertebral enumeration? A study of 3.0-Tesla lumbar MRI comparing skeletal, neural, and vascular markers, Clin. Imaging **38**, 792–796 (2014), doi:10.1016/j.clinimag.2014.05.001.

[6]J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and associated vertebral numeration, Eur. Spine J. **27**, 995–1004 (2018), doi:10.1007/s00586-018-5554-8.

[7]J. Hanhivaara, J. H. Määttä, J. Karppinen, J. Niinimäki, and M. T. Nevalainen, The association of lumbosacral transitional vertebrae with low back pain and lum-

bar degenerative findings in MRI: a large cohort study, Spine **47**, 153–162 (2022), doi:10.1097/BRS.0000000000004244.

[8]S. Nagata et al., Lumbosacral transitional vertebra on whole-spine CT: prevalence and association with rib abnormalities, Skeletal Radiol. **54**, 2169–2177 (2025), doi:10.1007/s00256-025-04952-z.

[9]J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, Radiol. Artif. Intell. **5**, e230024 (2023), doi:10.1148/ryai.230024.

[10]A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, Med. Image Anal. **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.

[11]H. Möller et al., SPINEPS—automatic whole spine segmentation of T2-weighted MR images using a two-phase approach to multi-class semantic and instance segmentation, Eur. Radiol. **35**, 1178–1189 (2025), doi:10.1007/s00330-024-11155-y.

[12]H. Möller et al., VERIDAH: Solving Enumeration Anomaly Aware Vertebra Labeling across Imaging Sequences, arXiv, 2026, arXiv:2601.14066. doi:10.48550/ARXIV.2601.14066.

[13]Y. Otake et al., Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery, Phys. Med. Biol. **57**, 5485–5508 (2012), doi:10.1088/0031-9155/57/17/5485.

[14]S.-f. L. Lo et al., Automatic Localization of Target Vertebrae in Spine Surgery: Clinical Evaluation of the LevelCheck Registration Algorithm, Spine (Phila. Pa. 1976) **40**, E476–E483 (2015), doi:10.1097/BRS.0000000000000814.

[15]T. De Silva et al., Utility of the LevelCheck Algorithm for Decision Support in Vertebral Localization, Spine (Phila. Pa. 1976) **41**, E1249–E1256 (2016), doi:10.1097/BRS.0000000000001589.

[16]T. A. Hutchins et al., ACR Appropriateness Criteria Low Back Pain: 2021 Update, J. Am. Coll. Radiol. **18**, S361–S379 (2021), doi:10.1016/j.jacr.2021.08.002.

[17]D. S. Kreiner et al., An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update), Spine J. **13**, 734–743 (2013), doi:10.1016/j.spinee.2012.11.059.

[18]American College of Radiology, ACR–ASNR–ASSR–SPR practice parameter for the performance of computed tomography (CT) of the spine, revised 2022 (Resolution

23), American College of Radiology, Reston, VA. https://gravitas.acr.org/PPTS/GetDocumentView?docId=133, 2022.

[19] American College of Radiology, ACR–ASNR–SABI–SSR practice parameter for the performance of magnetic resonance imaging (MRI) of the adult spine, revised 2023 (Resolution 6), American College of Radiology, Reston, VA. https://gravitas.acr.org/PPTS/GetDocumentView?docId=4, 2023.

[20] S. Sixta et al., Screening for thoracolumbar spinal injuries in blunt trauma: an Eastern Association for the Surgery of Trauma practice management guideline, J. Trauma Acute Care Surg. **73**, S326–S332 (2012), doi:10.1097/TA.0b013e31827559b8.

[21] American College of Surgeons Committee on Trauma, ACS TQIP best practices guidelines in imaging, American College of Surgeons, Chicago. https://www.facs.org/media/oxdjw5zj/imaging_guidelines.pdf, 2018.

[22] K. Smith et al., Data from CT COLONOGRAPHY (ACRIN 6664), The Cancer Imaging Archive, version 2 [dataset], 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

[23] K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, J. Digit. Imaging **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.

[24] P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, Int. J. Comput. Assist. Radiol. Surg. **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.

[25] Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, Mach. Learn. Biomed. Imaging **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.

[26] A. M. Du Plessis, L. M. Greyling, and B. J. Page, Differentiation and classification of thoracolumbar transitional vertebrae, J. Anat. **232**, 850–856 (2018), doi:10.1111/joa.12781.

[27] A. M. Poolman, Q. Wessels, A. V. Schoor, and N. Keough, Thoracolumbar transitional vertebrae: Quantitative differentiation and associated numeric variation in the vertebral column using skeletal remains, J. Anat. **243**, 311–318 (2023), doi:10.1111/joa.13865.

[28] M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and lumbar pedicles, Spine (Phila. Pa. 1976) **12**, 160–166 (1987), doi:10.1097/00007632-198703000-00012.

[29]M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy, Spine (Phila. Pa. 1976) **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.

[30]M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy, Spine (Phila. Pa. 1976) **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.

[31]J. L. Berry, J. M. Moran, W. S. Berg, and A. D. Steffee, A morphometric study of human lumbar and selected thoracic vertebrae, Spine (Phila. Pa. 1976) **12**, 362–367 (1987), doi:10.1097/00007632-198705000-00010.

[32]R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on MRI: role of the iliolumbar ligaments, AJR Am. J. Roentgenol. **187**, W59–W65 (2006), doi:10.2214/AJR.05.0415.

[33]D. Schinz et al., A vertebral shape-based classification for labeling the thoracolumbar junction on CT imaging, Eur. Spine J. **35**, 2195–2203 (2026), doi:10.1007/s00586-025-09364-0.

[34]H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, Sci. Data **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.

[35]L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, IEEE Trans. Med. Imaging **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.

[36]National Electrical Manufacturers Association, Digital Imaging and Communications in Medicine (DICOM) Standard, Part 3: Information Object Definitions, Technical Report PS3.3, NEMA, Rosslyn, VA, 2024, Attributes (0010,0020), (0020,000D), (0020,000E), (0018,5100).

[37]R. W. Cox, J. Ashburner, H. Breman, K. Fissell, C. Haselgrove, C. J. Holmes, J. L. Lancaster, D. E. Rex, S. M. Smith, J. B. Woodward, and S. C. Strother, A (sort of) new image data format standard: NIfTI-1, NeuroImage **22**, e1440 (2004), 10th Annual Meeting of the Organization for Human Brain Mapping, Budapest.

[38]X. Li, P. S. Morgan, J. Ashburner, J. Smith, and C. Rorden, The first step for neuroimaging data analysis: DICOM to NIfTI conversion, J. Neurosci. Methods **264**, 47–56 (2016), doi:10.1016/j.jneumeth.2016.03.001.

[39]K. Hasegawa, M. Okamoto, S. Hatsushikano, G. Caseiro, and K. Watanabe, Difference in whole spinal alignment between supine and standing positions in patients with adult spinal

deformity, BMC Musculoskelet. Disord. **19**, 437 (2018), doi:10.1186/s12891-018-2355-5.

[40]F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, Nat. Methods **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

[41]H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, arXiv, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

[42]A. Schehr, J. Kim, and G. Schwing, OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research, Cureus (2026), doi:10.7759/cureus.112661.

[43]N. J. Veilleux, N. V. Kalore, J. A. Vossen, and J. S. Wayne, Automatic characterization of pelvic and sacral measures from 200 subjects, J. Bone Joint Surg. Am. **102**, e130 (2020), doi:10.2106/JBJS.20.00343.

[44]C. M. Lee and R. W. Liu, Comparison of pelvic incidence measurement using lateral x-ray, standard CT versus CT with 3D reconstruction, Eur. Spine J. **31**, 241–247 (2022), doi:10.1007/s00586-021-07024-7.

[45]R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, J. Bone Joint Surg. Am. **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

[46]P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, Ann. Intern. Med. **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.

[47]M. G. Mody, A. Nourbakhsh, D. L. Stahl, M. Gibbs, M. Alfawareh, and K. J. Garges, The Prevalence of Wrong Level Surgery Among Spine Surgeons, Spine (Phila. Pa. 1976) **33**, 194–198 (2008), doi:10.1097/BRS.0b013e31816043d1.

[48]N. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, Surg. Neurol. Int. **12**, 286 (2021), doi:10.25259/SNI_402_2021.

[49]E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015, ISBN 978-1-60406-924-2.

[50]G. J. Hahn, W. Q. Meeker, and L. A. Escobar, *Statistical Intervals: A Guide for Practitioners and Researchers*, Wiley, Hoboken, NJ, 2nd edition, 2017.

[51]E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, J. Clin. Med. **15**, 2584 (2026), doi:10.3390/jcm15072584.

FIGURE CAPTIONS

Figure 1. How the dataset was built. n counts scans, or corrections, at each step.

Figure 2. Four phenotypes at the lumbar borders, aligned on the sacrum. Red, lowest rib-bearing vertebra and its rib; blue, sacrum; yellow, the lumbar bodies. Cases 0704, 0428, 0094, 0005, each the most ordinary member of its phenotype by rule. (b) and (c) present the same upper border, a short rib pair under a full pair, and were named differently.

Figure 3. Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile, n at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is Panjabi's cadaveric series[29,30], keyed in (a), its SD recovered from the reported SEM; T11–T12 EPWu is digitised from his Figure 4A.